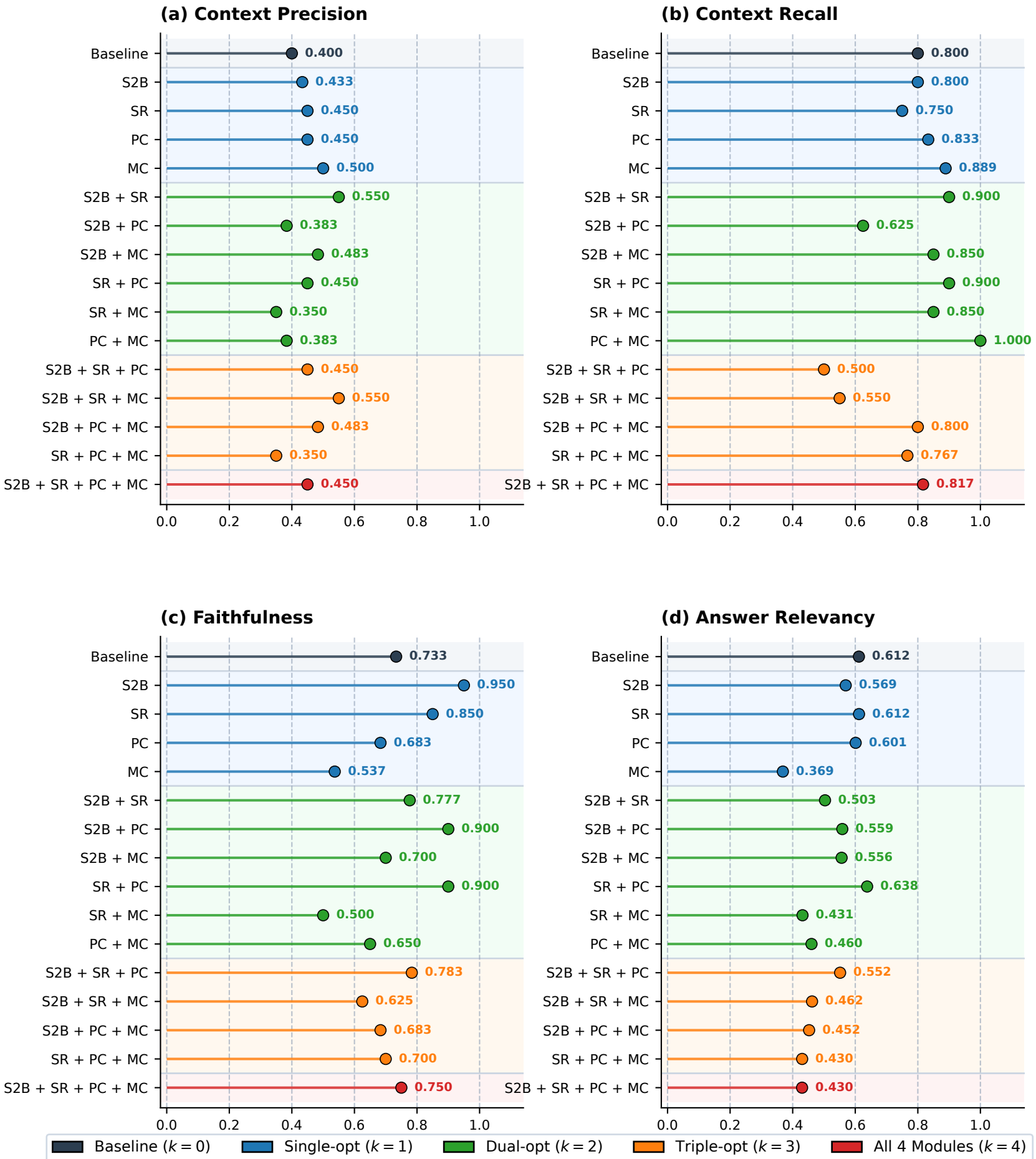
RAGAS Quality Metrics by Pipeline Architecture — MuSiQue
(a) Context Precision
Baseline 0.400
S2B 0.433
SR 0.450
PC 0.450
MC 0.500
S2B + SR 0.550
S2B + PC 0.383
S2B + MC 0.483
SR + PC 0.450
SR + MC 0.350
PC + MC 0.383
S2B + SR + PC 0.450
S2B + SR + MC 0.550
S2B + PC + MC 0.483
SR + PC + MC 0.350
S2B + SR + PC + MC 0.450
0.0 0.2 0.4 0.6 0.8 1.0
(b) Context Recall
Baseline 0.800
S2B 0.800
SR 0.750
PC 0.833
MC 0.889
S2B + SR 0.900
S2B + PC 0.625
S2B + MC 0.850
SR + PC 0.900
SR + MC 0.850
PC + MC 1.000
S2B + SR + PC 0.500
S2B + SR + MC 0.550
S2B + PC + MC 0.800
SR + PC + MC 0.767
S2B + SR + PC + MC 0.817
0.0 0.2 0.4 0.6 0.8 1.0
(c) Faithfulness
Baseline 0.733
S2B 0.950
SR 0.850
PC 0.683
MC 0.537
S2B + SR 0.777
S2B + PC 0.900
S2B + MC 0.700
SR + PC 0.900
SR + MC 0.500
PC + MC 0.650
S2B + SR + PC 0.783
S2B + SR + MC 0.625
S2B + PC + MC 0.683
SR + PC + MC 0.700
S2B + SR + PC + MC 0.750
0.0 0.2 0.4 0.6 0.8 1.0
(d) Answer Relevancy
Baseline 0.612
S2B 0.569
SR 0.612
PC 0.601
MC 0.369
S2B + SR 0.503
S2B + PC 0.559
S2B + MC 0.556
SR + PC 0.638
SR + MC 0.431
PC + MC 0.460
S2B + SR + PC 0.552
S2B + SR + MC 0.462
S2B + PC + MC 0.452
SR + PC + MC 0.430
S2B + SR + PC + MC 0.430
0.0 0.2 0.4 0.6 0.8 1.0
Baseline ($k = 0$)
Single-opt ($k = 1$)
Dual-opt ($k = 2$)
Triple-opt ($k = 3$)
All 4 Modules ($k = 4$)